

Automated Research & Blog Engine (ARBE)

RhoFi Financial Technologies – Intern Module

1. Executive Summary

The **Automated Research & Blog Engine (ARBE)** is a content-intelligence module designed to automatically generate **enriched, short, readable market insights** (1–3 minutes reading time) using RhoFi’s market data, internal analytics, and AI-driven narrative generation.

ARBE aims to transform raw financial signals into **daily insights, stock spotlights, MF market notes, and macro summaries**, reducing manual writing effort and enabling RhoFi to maintain a consistent publishing schedule.

This is **not** a traditional blog CMS.

It is a **full automation engine** that **detects signals → generates content → prepares drafts → allows human approval → publishes**.

Interns will **own the foundational build** of the automation system.

2. Business Problem

RhoFi currently produces education and market insights manually, which is:

- Time-consuming
- Dependent on author availability

- Difficult to scale
- Delays consistent content flows
- Limits RhoFi's ability to send daily/weekly insights

RhoFi requires a **lightweight-but-robust automated content engine** that can:

- Scan market movements
- Summarize trends
- Generate context-rich short articles
- Publish polished content quickly
- Maintain a consistent brand voice

3. Module Objectives

- ✓ **Automatically detect key market events**
- ✓ **Generate high-quality, enriched content summaries**
- ✓ **Convert insights into 1–3 min readable blogs**
- ✓ **Provide editor review workflow (human-in-the-loop)**
- ✓ **Publish to RhoFi Blog with scheduling & meta-tags**
- ✓ **Trigger notifications via in-app, email, WhatsApp**

4. User Personas

Persona	Role	Needs
Founder (Sekar)	Publisher	Daily consistent content without manual effort
Product Lead (Rajani)	Strategy	Reliable insights feed
Intern Developers	Engineers	Clear automation logic
Design Lead (Gopal)	UX	Clean reading layout
User (Reader)	Consumer	Simple, enriched 1–3 min insights

5. Functional Requirements

5.1 Data Acquisition Layer

Engine must gather data from:

Internal RhoFi Modules:

- RSA (Equity signals)
- MF Engine (returns, category trends)
- Market data (NIFTY, BN, sector moves)
- News summaries (phase-2)

Data must be cached for performance.

5.2 Signal Detection Logic (Intern-owned)

The system must auto-detect events like:

Market Signals:

- NIFTY movement $> \pm 0.8\%$
- BankNifty $> \pm 1\%$
- Sector rotation (top gainers/losers)

Stock Signals:

- Large daily move ($> 2-3\%$)

- Unusual volume
- Breakout/breakdown signals

MF Signals:

- Category outperformers
- SIP inflow trends
- Volatility protection funds rising

Macro Signals:

- RBI announcements
- Inflation/GDP updates
- Crude oil or USDINR volatility

Each signal triggers a content template.

5.3 AI Template-Based Content Generator

System must map signals → templates → generated drafts.

Content Types:

1. **Market Summary (1–2 min)**
2. **Stock Spotlight (1 min)**
3. **MF Insight Note (1–2 min)**
4. **Macro Update Summary (1 min)**
5. **RhoFi Quick Learning Snippets (30–60 sec)**
6. **Daily Morning Brief / Evening Wrap (2–3 min)**

Content Qualities Required:

- Enriched
 - Insightful
 - Simple finance language
 - Non-advisory tone
 - 1–3 minute reading time
 - Bullet points preferred for clarity
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5.4 Draft → Review → Publish Workflow

States:

- Draft (auto-generated)
- Under Review
- Approved
- Scheduled
- Published
- Archived

Admin actions allowed:

- Edit
 - Regenerate
 - Approve
 - Reject
 - Schedule publish
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5.5 Scheduling Engine

Admin can schedule:

- Immediate publish
- Timed publish (e.g., 8:30 AM “Morning Insight”)
- Recurring templates (phase-2)

Engine must use cron-like worker or queue processor.

5.6 SEO Metadata

Each generated article must include:

- Title
 - 150–200 char summary
 - Keywords
 - Category tags
 - Estimated reading time
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5.7 Publishing Pipeline

After approval:

- Article is published to public blog
 - Stored in `blog_posts` table
 - Notification event triggered
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5.8 Notification Trigger

When published:

- In-App notification created
- Email draft prepared
- WhatsApp template triggered (future)

Interns only prepare **payload format**, not integration.

6. Non-Functional Requirements

Category	Requirement
Content Quality	Enriched, simple, accurate
Performance	Content generation < 2 seconds
Scaling	Able to generate 50+ drafts/day
Security	Admin-only access
Compliance	No investment advice wording
Version Control	Draft history maintained

7. Data Model (DB Schema)

Table: blog_auto_signals

Field	Type	Description
signal_id	UUID	PK
signal_type	string	market/stock/mf/macro
payload_json	JSON	data snapshot
generated_at	timestamp	detected time

Table: blog_auto_drafts

Field	Type	Description
draft_id	UUID	PK
signal_id	UUID	FK
title	string	Generated title
summary	string	SEO summary
body_html	text	Generated content
status	string	draft/review/scheduled
scheduled_at	timestamp	optional
created_at	timestamp	timestamp

Table: blog_published

Field	Type	Description
post_id	UUID	PK
title	string	Final title
slug	string	URL slug
body_html	text	Final content
meta_tags	JSON	SEO metadata
category	string	mf/equity/market
published_at	timestamp	published date

8. API Specifications

POST /arbe/generate

Body:

```
{ "signal_type": "market_summary", "payload": {...} }
```

GET /arbe/drafts

POST /arbe/draft/{id}/approve

POST /arbe/draft/{id}/schedule

POST /arbe/draft/{id}/regenerate

GET /blogs/list

GET /blogs/{slug}

9. User Flow Diagram

RhoFi Data → Signal Engine → ARBE Templates → AI Draft → Admin Review → Approve & Publish → Notify Users

10. UI Wireframes

Interns will design:

- Auto-generated draft list
- Draft editor
- Preview mode
- Scheduling dialog
- Published blogs list

Gopal will support on design.

11. Dependencies

Area	Contact
Signal logic	Sekar & Rajani
Backend integration	Shafvan
Infra	Boopalan
UI/UX	Gopal
QA	All interns

12. Risks & Mitigation

Risk	Mitigation
Overly technical writing	Use simpler tone in prompt templates
Incorrect financial interpretation	Sekar & Rajani verify templates

Risk	Mitigation
Large-scale traffic	Caching summaries
LLM hallucination	Include strict prompt guardrails

13. Acceptance Criteria

- ✓ Generate enriched, short 1–3 min insights
 - ✓ Auto-create drafts with titles, summaries
 - ✓ Admin can edit, approve, publish
 - ✓ Scheduling engine works
 - ✓ SEO metadata generated
 - ✓ Notifications integrated (payload ready)
 - ✓ At least **5 types of automated insights working**
 - ✓ Demo delivery validated by Sekar & Rajani
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14. 1-Month Delivery Plan

Week 1

- Understand financial data, signals
- Build DB schema
- Create template library

Week 2

- Implement signal detection
- Auto-draft generation
- Draft list + preview

Week 3

- Scheduling engine
- Publishing pipeline
- SEO metadata

Week 4

- UI polishing
 - QA automation
 - Final demo to founder
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